

Cristian Ocampo-Padilla

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Education

University of Illinois Urbana-Champaign

Professional Master of Computer Science

Aug 2025 – Dec 2026

Chicago, IL

GPA: 4.00/4.00

Relevant Coursework: Computer Security, Cloud Computing, Machine Learning for SWE, IoT Systems

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science

May 2024

GPA: 3.67/4.00

Relevant Coursework: Data Structures & Algorithms, Systems Programming, Databases, Software Engineering, Parallel Programming, High-Frequency Trading Technology

Experience

International Motors (Navistar)

Lisle, IL

Software Engineer Co-Op

Aug 2023 – Dec 2023

- Gathered and reviewed system requirements to design test scenarios and validate results
- Executed and validated Python scripts to ensure accurate data transfer and correct outputs across systems
- Implemented error handling and logging to diagnose failures and support debugging
- Collaborated with engineers in an Agile environment to identify issues and improve reliability of internal tools
- Documented findings and reported issues to teammates for timely resolution

Projects

Inventory & Reservation Management Web App

React, FastAPI, MySQL, GCP

- Developed a full-stack web application to manage inventory and reservations for shared resources
- Designed and executed test cases to validate API endpoints, database operations, and user workflows
- Identified and debugged issues across frontend and backend components, improving system reliability
- Designed and tested MySQL database schema to ensure data integrity and correct handling of edge cases
- Collaborated with 2-3 peers in Agile-style development and testing cycles, documenting results for review

Enhanced GPS Grandmaster on Raspberry Pi 3

Ansible, Bash, Linux

- Tested and optimized time synchronization performance using CPU isolation and interrupt steering
- Collected and analyzed system-level timing data to diagnose performance inconsistencies
- Validated system accuracy using PPS signals, achieving ~100 nanosecond precision
- Debugged low-level system behavior and tuned CPU and thermal settings to improve stability
- Documented test results and configuration changes to ensure reproducibility

Skills

Languages: Python, C++, JavaScript, SQL, C

Testing & Debugging: Manual Testing, Test Case Design, Debugging, Log Analysis

Frameworks: React, FastAPI

Tools & Technologies: Git, GCP, AWS, Firebase, Ansible, Agile/Scrum

Systems: Linux, Windows

Languages: English (Fluent), Spanish (Fluent)